Ahmedabad University

PSY-310 Lab In Psychology

Guided Visual Search Report

Section-1

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GitHub Link - https://github.com/HetanshJalan/Psychology-Experiments

Introduction

Visual search is the process of finding a target in a visual scene among other items. Useful for studying attention as it required the observer to select and ignore information that is relevant or irrelevant to the target. Reaction time may thus provide a measure of the efficiency of the attentional flow through a display. As the number of distractors is increased, the reaction time increases and the extra items are inflicting a measurable cost on search.
Treisman and Gelade (1980) suggested that some basic visual features could be registered in a general way throughout an array of stimuli, whereas focused attention was required when the features needed to be combined into objects. Another factor in visual-search efficiency is the similarity of targets and non-targets: It is harder to search when the targets are more similar to the distractors (Duncan & Humphreys, 1989). In Guided Search, it is also suggested that information from prior visual processing guides attentional selection (Wolfe, 1994). Therefore, the slope of the reaction-time by set-size function is useful. The larger the positive slope, the more search time it will take to add each additional distractor; the flatter the slope, the more efficient the search over the set sizes tested.
Aim- To see if a larger set of distractors (5 vs. 10) would result in more time being taken to accurately select a target in a visual-search display.
Alternative Hypothesis - I would predict that the mean reaction time on correct trials would be faster in the 5 distractor condition than in the 10 distractor condition and have a positive slope for RT x set size.

Method

Participants

Data collected from five participants – Hetansh, Manthan, Pushti, Trusha and Vedanshi. A total of 1000 observations were made, with each participant making 200. These labels are the only demographic information included in the available data files and I have not included any other demographic information that did not appear in the available data files.

Materials

The experiment was designed and executed with PsychoPy 2026.2.3, a software package for controlling stimulus presentation and recording responses (Peirce et al., 2019). The refresh rate recorded from each of the data files was from about 59.67 to 60.07 Hz. The screen was grey, and had a white 'fixation cross' in the center. The target was a white letter T set at a 0.05 PsychoPy units letter height and Arial font. Distractors: 0.10 letter height white letter Ls with random rotation between 0 and 360 degrees. The location of target and distractors were randomly placed in the programmed display area. The participants used the mouse clicking the target.

Experiment design

The 200 trials were done using a random TrialHandler. It was determined at the start of each trial whether the code would pick 5 or 10 L distractors and it was saved as a variable called num_distr. Target T was then presented along with these distractors. Thus, the two programmed conditions had a total of 6 or 11

search items, but the independent variable stored in the data set was the number of distractors (5 or 10). A target was always in existence during each trial. The routine was terminated by a mouse click and PsychoPy recorded the component clicked and the response time.

Parameter	Setting
Software	PsychoPy 2026.2.3
Participants	5
Observations	200 per participant; 1,000 total
Target	White T; Arial; height 0.05
Distractors	White Ls; Open Sans; height 0.10
Distractor set sizes	5 or 10
Total visible search items	6 or 11, including the target
Distractor orientation	Random integer from 0° to 360°
Target / distractor position	Randomly generated within programmed display range
Response	Mouse click; target registered as clickable

Table 1 - Main visual-search experiment parameters.

Procedure

For each trial, I wanted to show one T target with either 5 or 10 distractors (L). The distractors were oriented randomly. The location of the target and distractors were randomly generated. The participant scanned the display and used the mouse to click on the T. PsychoPy saved the number of distractors, the component that was clicked and the mouse reaction time. This procedure was repeated for 200 trials for each of the five participants.

Data treatment

A target was classified as accurate when the mouse was clicked and its name was recorded as the target. As per assignment, only these "accurate" trials had reaction time calculated. The time taken by the mouse was converted to milliseconds. I determined the accurate-trial RT for each participant separately for 5 and 10 distractors. I then subtracted the mean RT at 5 distractors from the mean RT at 10 distractors and divided by 5 to get the individual search slope. The difference in time (in milliseconds) in the two set sizes tested is the resulting value.

Results

Overall accuracy was 77.4% (1000 observations, 774 correct) across all 5 participants. The mean RT for the accurate trials was 1,230.4 ms (TW). The mean RT for the combined data for accurate trials was 1,170.1 ms (5 distractors) and 1,288.2 ms (10 distractors).

Participant	Accurate	Accuracy	Mean accurate RT	RT: 5 distractors	RT: 10 distractors	Slope
Hetansh	141/200	70.5%	991.0 ms	912.2 ms	1040.0 ms	25.56 ms/distr.
Manthan	186/200	93.0%	1386.4 ms	1346.1 ms	1437.6 ms	18.30 ms/distr.
Pushti	167/200	83.5%	1465.6 ms	1422.7 ms	1502.2 ms	15.90 ms/distr.
Trusha	135/200	67.5%	1365.5 ms	1186.2 ms	1558.6 ms	74.48 ms/distr.
Vedanshi	145/200	72.5%	866.4 ms	833.1 ms	901.1 ms	13.61 ms/distr.

Table 2 - Mean RT of accurate trials and RT x set-size slope for each participant.

For my own data, 141 of 200 trials were accurate (70.5%). My mean RT across accurate trials was 991.0 ms. Mean accurate RT increased from 912.2 ms with 5 distractors to 1040.0 ms with 10 distractors. My RT x set-size slope was therefore 25.56 ms per distractor.

Every participant showed a positive slope. The individual slopes ranged from 13.61 to 74.48 ms per distractor, and the mean individual slope across the five participants was 29.57 ms per distractor. Thus, the larger distractor condition was associated with slower accurate search for all five participants.

Discussion

My hypothesis was supported by the results. All five participants had larger mean accurate-trial RT when 10 distractors than when 5 distractors were presented. My individual slope was 25.56 ms per distractor, and the mean individual slope was 29.57 ms per distractor. A positive slope indicates that the addition of distractors resulted in a cost in reaction time (RT) rather than no change in RT. In this experiment, the more competing L stimuli that were added to the display, the less efficient the attention was.

The amount of extra search time for extra distractors is related to attention, and is measured by the RT x set-size slope. The shallower the slope, the less the target will need to be selected with additional costs as the display increases, and the steeper the slope, the more it will be required to

be selected with additional costs as the display increases. This is in line with the general visual-search literature in which the efficiency of a search is influenced by the extent to which the visual information can capture the attention and by the similarity between targets and distractors (Duncan & Humphreys, 1989; Wolfe, 1994).

The result cannot be taken as an indication that the task was performed using a strictly sequential search procedure. Only two distractor counts were used in the experiment, which means that each slope was based on two RT means, not a detailed RT x set-size function. Also, the target was always there, and so target-absent search was not investigated. Furthermore, the letter shape and the size of the T target was different from the L distractors in the PsychoPy settings, which could have influenced attention. These design features restrict the extent of the inference that can be made about the exact nature of the search mechanism.

There was significant inter-individual variation, too. The steepest slope was shown by Trusha (74.48 ms/ distractor) and the shallowest slope was shown by Vedanshi (13.61 ms/ distractor). Accuracy also ranged from 67.5% to 93.0%. That is why it was important to have a task that asked each participant to compute a slope: the presence of a single group mean would have concealed the marked differences in search performance.

Conclusion

Overall accuracy was 77.4% for 1000 observations from the five subjects. The mean accurate-trial RT for the 5 distractors condition was 1170.1ms, and for the 10 distractors condition, 1288.2ms. All five single RT x set-size slopes were in the positive direction with a mean of 29.57 ms per distractor. The slope of my curve was 25.56ms per distractor. Increasing the number of distractors helped in this case to increase the accuracy of the target's localization, but it took longer time to perform the visual search task.

References

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